COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010	**SECTION:**	CPE21S1
DATE PERFORMED:	August 27, 2026	**DATE SUBMITTED:**	August 27, 2026
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PRELIM EXAM **[ACTIVITY NAME]**

1. PROCEDURE OUTPUT

Instructions:

- Please make sure to follow all code of conduct re: examinations and related policies.

Problem Title: The Palindrome Number Problem

Problem Definition:



A palindrome number is a number that remains the same when digits are reversed. For example, the number 12321 is a palindrome number, but 1451 is not a palindrome number. Given a positive integer, the task is to check whether the number is palindrome or not. We want to use data structures to check whether a given string of input conforms to the given description of a palindrome. The user is then prompted whether the given input is a palindrome or not.

Your task is to use two data structures for your solution:

- Stack
- Queue

But, they must have different internal representations, either:

- Array-based
- Linked List

Example: If your stack is array-based, your queue must be implemented using a linked list (and vice-versa).

You must then use the values in the given table below to validate your code.

Palindrome?	Output (Console)	Remark
10201		
1003003001		
88		
0		
202		

Pseudocode

```
START
Enter a number

Check if the input contains only digits
If not, display "Invalid input" and STOP

Put each digit into the Stack
Put each digit into the Queue

Compare the values from the Stack and Queue

If all values are the same
    Display "The number is a PALINDROME"
```

Else
    Display "The number is NOT a PALINDROME"
END

Analysis for Pseudocode
The program puts the digits into a Stack and a Queue. The Stack gives the digits in reverse order, while the Queue gives them in the original order. If the values are the same when compared, the number is a palindrome.

stackarray.h

```
#ifndef STACK_ARRAY_H
#define STACK_ARRAY_H

#include <iostream>

const int MAX = 100;

// Stack array
char stack[MAX];
int top = -1;

// Push
void push(char data)
{
    if (top == MAX - 1)
    {
        std::cout << "Stack Overflow" << std::endl;
        return;
    }

    top++;
    stack[top] = data;
}
```

```
// Pop
char pop()
{
    if (top == -1)
    {
        std::cout << "Stack Underflow" << std::endl;
        return '\0';
    }

    char value = stack[top];
    top--;

    return value;
}


// Is Empty
bool isEmpty()
{
    return top == -1;
}

#endif
```

queuell.h

```
#ifndef QUEUELL_H
#define QUEUELL_H

#include <iostream>

template <typename T>
class qNode
{
public:
    T data;
```

```
    qNode<T>* next;
};


// Create New Node
template <typename T>
qNode<T>* new_node(T newData)
{
    qNode<T>* newNode = new qNode<T>;

    newNode->data = newData;
    newNode->next = nullptr;

    return newNode;
}


// Enqueue
template <typename T>
void enqueue(qNode<T>** frontPtr, qNode<T>** backPtr, T newData)
{
    qNode<T>* newNode = new_node(newData);

    if (*frontPtr == nullptr && *backPtr == nullptr)
    {
        *frontPtr = newNode;
        *backPtr = newNode;
        return;
    }

    (*backPtr)->next = newNode;
    *backPtr = newNode;
}


// Dequeue
template <typename T>
```

```
T dequeue(qNode<T>** frontPtr, qNode<T>** backPtr)
{
    if (*frontPtr == nullptr)
    {
        return '\0';
    }

    qNode<T>* deleteNode = *frontPtr;

    T value = deleteNode->data;

    if ((*frontPtr)->next == nullptr)
    {
        *frontPtr = nullptr;
        *backPtr = nullptr;
    }
    else
    {
        *frontPtr = (*frontPtr)->next;
    }

    delete deleteNode;

    return value;
}

#endif
```

main.cpp

```
#include <iostream>
#include <string>

#include "stackarray.h"
#include "queuell.h"
```

```cpp
// Check Palindrome
bool checkPalindrome(std::string number)
{
    // Reset stack
    top = -1;

    // Queue pointers
    qNode<char>* front = nullptr;
    qNode<char>* back = nullptr;

    // Store values in Stack and Queue
    for (int i = 0; i < number.length(); i++)
    {
        push(number[i]);
        enqueue(&front, &back, number[i]);
    }

    // Compare Stack and Queue
    for (int i = 0; i < number.length(); i++)
    {
        char stackValue = pop();
        char queueValue = dequeue(&front, &back);

        if (stackValue != queueValue)
        {
            return false;
        }
    }

    return true;
}

int main()
{
    std::string number;

    std::cout << "================================" << std::endl;
```

```
    std::cout << "     Palindrome Number Check" << std::endl;
    std::cout << "==============================" << std::endl;

    std::cout << "Enter a number: ";
    std::cin >> number;

    // Check if input contains only digits
    for (int i = 0; i < number.length(); i++)
    {
        if (number[i] < '0' || number[i] > '9')
        {
            std::cout << "\nInvalid input. Please enter an integer."
                      << std::endl;
            return 0;
        }
    }

    if (checkPalindrome(number))
    {
        std::cout << "\nThe number is a PALINDROME." << std::endl;
    }
    else
    {
        std::cout << "\nThe number is NOT a PALINDROME." << std::endl;
    }

    return 0;
}
```

Code Analysis

The code uses an array-based Stack and a linked-list Queue to check if a number is a palindrome. Each digit is stored in both data structures, then the Stack and Queue remove their values in different orders for comparison. The program also checks if the input contains only numbers and displays an error for invalid input. If all the compared values are the same, the program displays that the number is a palindrome.

Palindrome	Output (Console)	Remarks
10201	========== Palindrome Number Check ========== Enter a number: 10201 The number is a PALINDROME.	Yes
1003003001	========== Palindrome Number Check ========== Enter a number: 1003003001 The number is a PALINDROME.	Yes
88	========== Palindrome Number Check ========== Enter a number: 88 The number is a PALINDROME.	Yes
0	========== Palindrome Number Check ========== Enter a number: 0 The number is a PALINDROME.	Yes
202	========== Palindrome Number Check ========== Enter a number: 202 The number is a PALINDROME.	Yes

Conclusion/Analysis
In this practical, I learned how to use a Stack and Queue to check if a number is a palindrome. The Stack stores the digits in reverse order, while the Queue keeps the digits in their original order. By comparing the values from both data structures, the program can determine if the number is the same forwards and backwards. I also learned how to use an array-based Stack and a linked-list Queue in one program.

Overall, this practice helped me understand how Stack and Queue work and how they can be used together to solve a problem. I learned that the Stack uses LIFO while the Queue uses FIFO. The activity also improved my understanding of linked lists, arrays, and comparing data using C++.

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

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